Progressive sharpening from the dynamics of the error

Draft section (not yet part of main.tex)

Draft

This note asks why the largest eigenvalue of the loss Hessian grows during training (progressive sharpening), and why it grows more for rougher targets. It uses the objects of main.tex: sampled functions with $\langle a, b\rangle = \frac{1}{n}\sum_i a_i \cdot b_i$, the gradient features Ψ, the training kernel $K = \frac{1}{n}\Psi\Psi^\top$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ and eigenvectors v_k, the error $E = \hat{Y} - Y$ with components $\hat{e}_k = \langle v_k, E\rangle$, and the slow part $E_\gamma = \gamma(K + \gamma I)^{-1}E$. Throughout, $\Pi = I$ and $c_p = 1$, the loss is $L = \frac{1}{2}\|E\|^2$, its gradient is $g = \frac{1}{n}\Psi^\top E$, gradient descent is $\theta_{k+1} = \theta_k - \eta g$, and gradient flow is $\dot{\theta} = -g$, under which $\dot{E} = -KE$.

Every statement below is an identity or an inequality with all terms kept. Where a statement is made for gradient flow, the corresponding statement for one step of gradient descent is given with its remainder written out. Section 8 lists what is proved, what is proved only with the kernel held fixed, and what is measured.

1 The loss Hessian and the kernel

For a sampled function φ, let $S(\varphi) = \frac{1}{n}\sum_i \varphi_i \cdot \nabla^2_\theta \mathcal{N}_\theta(x_i)$ be the Hessian in the parameters of $\langle \varphi, \mathcal{N}_\theta\rangle$, and let $g_\varphi = \frac{1}{n}\Psi^\top\varphi$, so that $\langle \varphi, K\varphi\rangle = |g_\varphi|^2$.

Proposition 1. *At every θ at which $\mathcal{N}_\theta$ is twice differentiable at the training inputs,*

$$\nabla^2 L = G + S(E), \qquad G = \frac{1}{n}\Psi^\top\Psi. \tag{1}$$

G and K have the same nonzero eigenvalues, and

$$\big|\lambda_{\max}(\nabla^2 L) - \lambda_1(K)\big| \leq \|S(E)\|_{op} \leq M_2\,\|E\|, \qquad M_2 = \Big(\tfrac{1}{n}\textstyle\sum_i \|\nabla^2_\theta \mathcal{N}_\theta(x_i)\|^2_{op}\Big)^{1/2}. \tag{2}$$

Proof. $L = \frac{1}{2n}\sum_i |\mathcal{N}_\theta(x_i) - Y_i|^2$ is quadratic in the outputs, so differentiating twice gives Eq. (1) with no remainder. $\frac{1}{n}\Psi^\top\Psi$ and $\frac{1}{n}\Psi\Psi^\top$ share their nonzero eigenvalues. Weyl's inequality gives the first inequality of Eq. (2), and $\|S(E)\|_{op} \leq \frac{1}{n}\sum_i |E_i|\,\|\nabla^2\mathcal{N}_\theta(x_i)\|_{op}$ with the Cauchy–Schwarz inequality gives the second. □

At an interpolating point, $E = 0$, the two sharpnesses are equal. The rest of the note studies $\lambda_1(K)$.

2 The change of λ_1 in one step

Proposition 2. *Let $\lambda_1(\theta) = \lambda_1(K(\theta))$, and let $\theta' = \theta - \eta g$.*

(a) If Ψ is locally Lipschitz in θ, then λ_1 is locally Lipschitz along the segment from θ to θ', it is differentiable at almost every point of the segment, and

$$\lambda_1(\theta') - \lambda_1(\theta) = -\eta \int_0^1 \nabla\lambda_1(\theta - \tau\eta g) \cdot g \, d\tau. \tag{3}$$

(b) At a point where λ_1 is a simple eigenvalue and Ψ is differentiable, $\nabla\lambda_1 = 2\, S(v_1)\, g_{v_1}$, with $\|v_1\| = 1$.

(c) Write $h = S(v_1)g_{v_1}$ and $q = \Psi h$, the change of the output produced by the parameter direction h. Then

$$-\eta\, \nabla\lambda_1 \cdot g = -2\eta\, \langle q, E\rangle = -2\eta \sum_k \hat{q}_k\, \hat{e}_k, \qquad \hat{q}_k = \langle v_k, q\rangle, \tag{4}$$

and Eq. (3) reads $\lambda_1(\theta') - \lambda_1(\theta) = -2\eta\langle q, E\rangle + R$ with

$$R = -\eta \int_0^1 \big(\nabla\lambda_1(\theta - \tau\eta g) - \nabla\lambda_1(\theta)\big) \cdot g \, d\tau. \tag{5}$$

If λ_1 is twice continuously differentiable on the segment, $|R| \le \frac{1}{2}\eta^2 |g|^2 \max_\tau \|\nabla^2\lambda_1(\theta - \tau\eta g)\|_{op}$.

Proof. (a) By Weyl's inequality, $|\lambda_1(K) - \lambda_1(K')| \le \|K - K'\|_{op}$, and K is locally Lipschitz when Ψ is. A Lipschitz function of one variable is absolutely continuous, so it is differentiable almost everywhere and is the integral of its derivative (Rademacher's theorem and the fundamental theorem of calculus for absolutely continuous functions). (b) $\lambda_1 = \max_{\|v\|=1}\langle v, Kv\rangle = |g_{v_1}|^2$. For a simple eigenvalue the maximizer is differentiable, and its variation does not change the value to first order, so $\nabla\lambda_1 = \nabla_\theta |g_v|^2\big|_{v=v_1} = 2\,(\nabla_\theta g_v)^\top g_v = 2\, S(v_1) g_{v_1}$, since $g_v = \nabla_\theta\langle v, \mathcal{N}_\theta\rangle$. (c) $\nabla\lambda_1 \cdot g = 2h \cdot \frac{1}{n}\Psi^\top E = 2\langle \Psi h, E\rangle$; expand in the eigenbasis of K. The remainder is Eq. (3) minus its value at $\tau = 0$; the bound is Taylor's theorem with integral remainder. □

Eq. (4) assigns to each eigendirection of K its exact share of the first-order change of λ_1, without dividing by λ_k. The direction of the change is set by q, which is built from the second derivatives of the network; the kernel alone does not determine it. A model that is linear in its parameters has $S \equiv 0$, so $q = 0$: its kernel never changes, whatever the target. Any explanation of sharpening therefore needs a property of the second derivatives of the network. Section 4 gives an exact decomposition, for any activation, that isolates the part of q along the output.

3 Learning speed and the loss Hessian

Let $\rho = \langle E, KE\rangle / \|E\|^2$. By part (c) of the proposition "Rates at one training time" of main.tex, gradient flow decreases $\|E\|$ at the rate $\rho\|E\|$. Let $\mathrm{Var}_E(\lambda) = \sum_k \lambda_k^2 w_k - \rho^2$ with weights $w_k = \hat{e}_k^2 / \|E\|^2$, the spread of the eigenvalues on which the error currently lies.

Proposition 3. *Under gradient flow, at a point where $\mathcal{N}_\theta$ is twice differentiable and $E \neq 0$, $g \neq 0$,*

$$\frac{d\ln\rho}{dt} = -\frac{2\,\mathrm{Var}_E(\lambda)}{\rho} + \frac{\langle E, \dot{K}E\rangle}{\langle E, KE\rangle}, \qquad \langle E, \dot{K}E\rangle = -2\, g^\top S(E)\, g, \tag{6}$$

$$\frac{g^\top \nabla^2 L\, g}{|g|^2} = \rho - \frac{1}{2}\frac{d\ln\rho}{dt} = \underbrace{\rho + \frac{\mathrm{Var}_E(\lambda)}{\rho}}_{G \text{ part}} + \underbrace{\frac{g^\top S(E) g}{|g|^2}}_{S(E) \text{ part}}. \tag{7}$$

Proof. $\langle E, KE\rangle = |g|^2$ and $\|E\|^2 = 2L$. Along gradient flow, $\dot L = -|g|^2$ and $\dot g = \nabla^2 L\,\dot\theta = -\nabla^2 L\, g$, so $\frac{d}{dt}\ln\rho = -2g^\top\nabla^2 L g/|g|^2 + 2|g|^2/\|E\|^2$, which is the first equality of Eq. (7). With Eq. (1), $g^\top G g = \|\Psi g\|^2 = \|KE\|^2$ and $\|KE\|^2/\langle E, KE\rangle = \rho + \mathrm{Var}_E(\lambda)/\rho$, which gives the second equality of Eq. (7). Eq. (6) follows by subtracting, and $\langle E, \dot K E\rangle = 2g\cdot\frac{1}{n}\dot\Psi^\top E = 2g\cdot S(E)\dot\theta$. □

With the kernel fixed, the learning speed ρ can only fall, at a rate set by the spread of the eigenvalues that carry the error; by the theorem "Rough errors are slow: the floor" of main.tex, a rough target leaves its error on the small eigenvalues. The only term that can raise ρ is the change of the kernel along the error, and Eq. (7) identifies it with the part $S(E)$ of the loss Hessian beyond G. For one step of gradient descent, the averaged-feature identity of main.tex (Section "The best roughness function") gives the exact change of the error, $E' = E - \eta\,\frac{1}{n}\bar\Psi\Psi^\top E$, with $\bar\Psi$ the features averaged along the step.

4 Layer scaling in any network

Consider a fully connected network with layers $l = 1, \dots, L$, pre-activations $z_l = W_l a_{l-1} + b_l$, activations $a_l = \sigma(z_l)$ for $l < L$, input $a_0 = x$, and a linear output layer, $\mathcal{N}_\theta = z_L$. The activation σ is any continuously differentiable function (twice, where S appears). Group the parameters by layer, $\theta_l = (W_l, b_l)$, write Ψ_l for the columns of Ψ that belong to layer l, $K_l = \frac{1}{n}\Psi_l\Psi_l^\top$, so that $K = \sum_l K_l$, $k_l = \langle v_1, K_l v_1\rangle$, and $g_l = \frac{1}{n}\Psi_l^\top E$. The output change produced by scaling layer l, and its deviation from the output, are the sampled functions

$$u_l = \Psi_l\theta_l, \qquad \delta_l = u_l - \hat Y, \tag{8}$$

and the deviation of the activation from homogeneity is $\psi(z) = z\,\sigma'(z) - \sigma(z)$. For ReLU, leaky ReLU and linear activations $\psi \equiv 0$; for tanh, $\psi(z) = z\,\mathrm{sech}^2 z - \tanh z = -\frac{2}{3}z^3 + O(z^5)$.

Lemma 1 (Layer scaling)**.** *At every point where $\mathcal{N}_\theta$ is differentiable at the training inputs:*

(a) $u_l(x) = \sum_j \frac{\partial \mathcal{N}_\theta}{\partial z_{l,j}}(x)\, z_{l,j}(x)$. *For the output layer, $u_L = \hat Y$ and $\delta_L = 0$.*

(b) For $l < L$,

$$\delta_l = \sum_{m=l}^{L-1}\Big(\sum_j \frac{\partial \mathcal{N}_\theta}{\partial a_{m,j}}\,\psi(z_{m,j}) - \sum_k \frac{\partial \mathcal{N}_\theta}{\partial z_{m+1,k}}\,b_{m+1,k}\Big), \tag{9}$$

the deviation of the activations from homogeneity in layers l to $L-1$, minus the contribution of the biases of layers $l+1$ to L.

(c) If λ_1 is simple and $\mathcal{N}_\theta$ is twice differentiable, $\theta_l \cdot h_l = \lambda_1 - k_l + \varepsilon_l$ with $\varepsilon_l = g_{v_1}\cdot\nabla_\theta\langle v_1, \delta_l\rangle$, where $h = S(v_1)g_{v_1}$ as in Proposition 2. For the output layer, $\varepsilon_L = 0$.

(d) One step of gradient descent changes the size of layer l by

$$|\theta_l'|^2 - |\theta_l|^2 = 2\eta\,\langle \hat Y + \delta_l, Y - \hat Y\rangle + \eta^2|g_l|^2. \tag{10}$$

Under gradient flow, $\frac{d}{dt}|\theta_l|^2 = 2\langle \hat Y + \delta_l, Y - \hat Y\rangle$; for the output layer, $\frac{d}{dt}|\theta_L|^2 = 2\langle \hat Y, Y - \hat Y\rangle$.

(e) Replacing θ_L by $c\,\theta_L$ multiplies K_m by c^2 for every $m < L$ and leaves K_L unchanged.

Proof. (a) $\partial\mathcal{N}_\theta/\partial W_{l,jk} = \frac{\partial\mathcal{N}_\theta}{\partial z_{l,j}} a_{l-1,k}$ and $\partial\mathcal{N}_\theta/\partial b_{l,j} = \frac{\partial\mathcal{N}_\theta}{\partial z_{l,j}}$, so $\Psi_l\theta_l = \sum_j \frac{\partial\mathcal{N}_\theta}{\partial z_{l,j}}\big(\sum_k W_{l,jk}a_{l-1,k} + b_{l,j}\big)$. For $l = L$, $\partial\mathcal{N}_\theta/\partial z_L = 1$. (b) With $\frac{\partial\mathcal{N}_\theta}{\partial a_{l,j}} = \sum_k \frac{\partial\mathcal{N}_\theta}{\partial z_{l+1,k}} W_{l+1,kj}$, part (a) for layer $l+1$ gives $u_{l+1} = \sum_j \frac{\partial\mathcal{N}_\theta}{\partial a_{l,j}}\sigma(z_{l,j}) + \sum_k \frac{\partial\mathcal{N}_\theta}{\partial z_{l+1,k}} b_{l+1,k}$, and part (a) for layer l gives $u_l = \sum_j \frac{\partial\mathcal{N}_\theta}{\partial a_{l,j}}\sigma'(z_{l,j})z_{l,j}$. Subtracting gives $u_l - u_{l+1}$ as the summand of Eq. (9); summing from l to $L-1$ and using $u_L = \hat{Y}$ gives Eq. (9). (c) Let P_l keep the coordinates of layer l. The function $\theta \mapsto g_{v_1}(\theta)\cdot P_l\theta = \langle v_1, u_l\rangle$ has gradient $S(v_1)P_l\theta + P_l g_{v_1}$, so $S(v_1)\theta_l = \nabla_\theta\langle v_1, u_l\rangle - P_l g_{v_1}$, with θ_l embedded in the full parameter vector. Since $S(v_1)$ is symmetric, $\theta_l \cdot h_l = g_{v_1}\cdot S(v_1)\theta_l = g_{v_1}\cdot\nabla_\theta\langle v_1, \hat{Y} + \delta_l\rangle - |P_l g_{v_1}|^2$, and $g_{v_1}\cdot\nabla_\theta\langle v_1,\hat{Y}\rangle = |g_{v_1}|^2 = \lambda_1$, $|P_l g_{v_1}|^2 = k_l$. (d) $|\theta_l - \eta g_l|^2 = |\theta_l|^2 - 2\eta\,\theta_l\cdot g_l + \eta^2|g_l|^2$ and $\theta_l\cdot g_l = \langle u_l, E\rangle$. (e) For $m < L$, $\partial\mathcal{N}_\theta/\partial\theta_m = W_L\,\partial a_{L-1}/\partial\theta_m$ is linear in W_L and does not involve b_L; $\partial\mathcal{N}_\theta/\partial\theta_L = (a_{L-1}, 1)$ does not depend on θ_L. □

When $\psi \equiv 0$ and there are no biases after the first layer, $\delta_l = 0$ and $\varepsilon_l = 0$ for every layer; then part (a) is Euler's identity for a function homogeneous of degree one in each layer, and part (d) is the balancedness of homogeneous networks of [7]. The function ψ is the "homogeneity defect" of [23], who use it, for weights without biases, to write the change of the difference of layer norms for any smooth activation; part (b) adds the biases and expresses the defect at the level of the output. For the output layer, part (d) is Eq. (2) of [17]. For tanh, δ_l comes from the cubic deviation ψ, which is small for small pre-activations and large where units saturate. The output layer is exact for every activation: its scaling reproduces the output, and it scales the kernel of every other layer.

Theorem 1 (Sharpening from the residual along the output)**.** *At a point where λ_1 is simple and $\mathcal{N}_\theta$ is twice differentiable, decompose $h = \sum_l c_l\theta_l + h_\perp$ with $c_l = \theta_l\cdot h_l/|\theta_l|^2$, the projection on the layer directions θ_l (orthogonal, since the layers are disjoint) plus a remainder. Then, exactly,*

$$-2\langle q, E\rangle = 2\alpha\,\langle\hat{Y}, Y - \hat{Y}\rangle + D + T, \qquad \alpha = \sum_l \frac{\lambda_1 - k_l}{|\theta_l|^2} \;\geq 0, \tag{11}$$

$$D = 2\Big(\sum_l \frac{\varepsilon_l}{|\theta_l|^2}\Big)\langle\hat{Y}, Y - \hat{Y}\rangle - 2\sum_l c_l\langle\delta_l, E\rangle, \qquad T = -2\langle\Psi h_\perp, E\rangle. \tag{12}$$

Keeping only the output layer in the projection gives, for every activation and with no defect term,

$$-2\langle q, E\rangle = 2\,\frac{\lambda_1 - k_L}{|\theta_L|^2}\,\langle\hat{Y}, Y - \hat{Y}\rangle + T_L, \qquad \frac{d\lambda_1}{dt} = \big(\lambda_1 - k_L\big)\,\frac{d\ln|\theta_L|^2}{dt} + T_L \;\; \textit{(gradient flow)}, \tag{13}$$

with $T_L = -2\langle\Psi(h - c_L\theta_L), E\rangle$. For one step of gradient descent, $\lambda_1(\theta') - \lambda_1(\theta) = \eta\,(-2\langle q, E\rangle) + R$ with R from Eq. (5).

Proof. $\Psi h = \sum_l c_l u_l + \Psi h_\perp = \big(\sum_l c_l\big)\hat{Y} + \sum_l c_l\delta_l + \Psi h_\perp$ by Eq. (8), and $\sum_l c_l = \alpha + \sum_l \varepsilon_l/|\theta_l|^2$ by Lemma 1(c). Eq. (4) gives Eqs. (11) and (12). Each $k_l \leq \sum_m k_m = \lambda_1$, so $\alpha \geq 0$. For the output layer, $\delta_L = 0$ and $\varepsilon_L = 0$, so the same computation with the single direction θ_L gives the first identity of Eq. (13), and Lemma 1(d) gives the second. □

In words: scaling the output layer reproduces the output, and it multiplies the kernel of all earlier layers, whose part along v_1 is $\lambda_1 - k_L = \sum_{m<L}\langle v_1, K_m v_1\rangle$. Gradient descent grows the output layer at the rate $2\langle\hat{Y}, Y - \hat{Y}\rangle$, the component of the residual along the output, so this term raises λ_1 whenever the output falls short of the target along itself. The hidden layers enter through $\lambda_1 - k_L$, through their own scaling terms and the defect D in Eq. (11), and through T, which collects the parameter moves that rescale no layer. The theorem does not bound T; Section 8 reports its measured size. [17] observed that sharpness and the output-layer norm move

together and took this as an assumption (their Assumption 3.1), with a proof for two-layer linear networks; Eq. (13) states the relation as an identity for any network with a linear output layer, with the coefficient $\lambda_1 - k_L$ and the remainder T_L written out. The general rate $-2\langle q, E\rangle$ is the top-eigenvalue contraction of the kernel evolution equation of the neural tangent hierarchy [10], and its sign is the progressive-sharpening coefficient of [6].

Proposition 4 (Size of the output and sharpness)**.** *For every activation and every layer l,*

$$\lambda_1(K) \;\geq\; \lambda_1(K_l) \;\geq\; \frac{\|u_l\|^2}{|\theta_l|^2}, \qquad \textit{in particular} \qquad \lambda_1(K) \;\geq\; \frac{\|\hat{Y}\|^2}{|\theta_L|^2}. \tag{14}$$

Proof. $K - K_l$ is positive semidefinite, so $\lambda_1(K) \geq \lambda_1(K_l)$, and $\|u_l\|^2 = \|\Psi_l\theta_l\|^2 = \theta_l^\top \frac{1}{n}\Psi_l^\top\Psi_l\theta_l \leq \lambda_1(K_l)|\theta_l|^2$. □

A roughness function gives nothing more here: since $\mathcal{R}(u) \leq C_{\mathcal{R}}\|u\|$ for every roughness function, the bound $\mathcal{R}(\hat{Y})^2/(C_{\mathcal{R}}^2|\theta_L|^2)$ of an earlier draft is never larger than Eq. (14), so the proposition does not distinguish rough outputs from smooth ones. Bounds of the same kind appear for linear chains in [27] and, through the input–output Jacobian, in [20, 19].

5 Where the residual along the output lies

In the eigenbasis of K, $\langle \hat{Y}, Y - \hat{Y}\rangle = -\sum_k \hat{y}_k\,\hat{e}_k$ with $\hat{y}_k = \langle v_k, \hat{Y}\rangle$. Write $y_k = \langle v_k, Y\rangle$.

Proposition 5 (Fixed kernel)**.** *Hold K fixed and start from $\hat{Y} = 0$.*

(a) Under gradient flow, $\hat{y}_k(t) = s_k(t)\,y_k$ with $s_k = 1 - e^{-\lambda_k t}$, and $\langle \hat{Y}, Y - \hat{Y}\rangle = \sum_k s_k(1-s_k)\,y_k^2 \geq 0$. Direction k contributes most when $\lambda_k t = \ln 2$, and $\int_0^\infty \langle \hat{Y}, Y - \hat{Y}\rangle\,dt = \frac{1}{2}\sum_{\lambda_k>0} y_k^2/\lambda_k$.

(b) After N steps of gradient descent, $s_k = 1 - (1-\eta\lambda_k)^N$. The contribution of direction k is nonnegative if $\eta\lambda_k \leq 1$, and alternates in sign with N if $1 < \eta\lambda_k < 2$.

(c) For the output $\hat{Y} = Y - Y_\gamma$, the smallest error reachable from zero by a step of length s_γ (main.tex, Proposition "The floor is a one-step bound"), $\langle \hat{Y}, Y - \hat{Y}\rangle = \gamma\,\Upsilon_Y(\gamma) = \gamma s_\gamma^2$.

Proof. (a) and (b): with K fixed, $\dot{E} = -KE$ or $E_N = (I - \eta K)^N E_0$ with $E_0 = -Y$; then $\hat{y}_k\hat{e}_k = -s_k(1-s_k)y_k^2$. The integral is $\int_0^\infty (e^{-\lambda t} - e^{-2\lambda t})dt = 1/(2\lambda)$. (c) $\hat{y}_k = \frac{\lambda_k}{\lambda_k+\gamma}y_k$ and $y_k - \hat{y}_k = \frac{\gamma}{\lambda_k+\gamma}y_k$, so the sum is $\gamma\sum_k \lambda_k y_k^2/(\lambda_k+\gamma)^2 = \gamma\Upsilon_Y(\gamma)$. □

Part (a) is Lemma 3.1 of [17], which also allows the eigenvalues to change with time when the eigenvectors are fixed. With Theorem 1, this gives the mechanism the note proposes: a direction that is already fitted ($s_k \approx 1$) or not yet started ($s_k \approx 0$) does not drive sharpening; a direction that is partly fitted does. Fast directions are fitted early, so late in training the drive comes from the slow directions, and a rough target, which by the floor theorem of main.tex puts its weight there, keeps the drive positive for a time of order $1/\lambda_k$ per direction and for a total that grows with $\sum_k y_k^2/\lambda_k$. Part (b) is where the edge of stability enters: once $\eta\lambda_1 > 1$, the top direction's contribution changes sign from step to step. Part (a) holds only with the kernel fixed; whether the residual along the output stays positive while the kernel moves is not proved here.

6 At the edge

This section concerns steps with $\eta\lambda_1 \approx 2$. It separates what gradient descent cannot do there, which is a statement with the features held fixed during the step, from what keeps λ_1 from growing further, which comes from the change of the features during the step.

Gradient descent and the best step of the same length.

Proposition 6. *Hold the features fixed during the step. The gradient-descent step has rescaled length $s = \eta\|E\|_K$ and leaves the error $(I-\eta K)E$, which keeps the fraction $1-\eta\lambda_k$ of the component of E along v_k. The best step of the same length leaves $E_\gamma = \gamma(K+\gamma I)^{-1}E$ with $\Upsilon(\gamma) = s^2$ (main.tex, Theorem "Supremum over roughness functions"(c)), which keeps the fraction $\gamma/(\lambda_k+\gamma) \in (0,1]$. Hence $\|(I-\eta K)E\| \geq F(s) = \|E_\gamma\|$, and*

$$\|(I-\eta K)E\|^2 - F(s)^2 = \sum_k \Big((1-\eta\lambda_k)^2 - \frac{\gamma^2}{(\lambda_k+\gamma)^2}\Big)\hat{e}_k^2. \tag{15}$$

Proof. To first order with the features fixed, the step changes the error by $-\eta\Psi g_E = -\eta KE$, and $|\eta g_E| = \eta\|E\|_K$. The best step is part (c) of the cited theorem, and the gradient step is one step of length s, so its error is at least $F(s)$. Eq. (15) is the difference of the two sums of squares in the eigenbasis. □

This is a restatement of classical facts in the notation of main.tex. The best step of a given length for a least-squares problem is the trust-region, or Levenberg–Marquardt, step [32, 33, 34], whose linearized residual is $\gamma(K+\gamma I)^{-1}E$ [30]; $E_\gamma = (I+K/\gamma)^{-1}E$ is the linearized implicit (proximal) step of gradient flow with step $1/\gamma$; and the best step lies in the Krylov space $\mathrm{span}\{E, KE, K^2E, \dots\}$, which conjugate-gradient trust-region methods search [35, 36], while gradient descent uses only KE with a fixed coefficient. At $\eta\lambda_1 = 2$ the gradient step keeps the fraction -1 of the component along v_1: it reverses it without reducing it, while a step of the same length can remove most of it. That the top component is multiplied by $1-\eta\lambda_1 \approx -1$ is the definition of the edge of stability [4], and reading it as the instability of an explicit scheme is in [26]. What Eq. (15) adds is a measurement of the per-step loss against the best step of the same length, split by eigendirection. Two cautions from the literature: projecting the top Hessian directions out of the update does as well as the full update [25], consistent with the top component making no progress; and step sizes that are greedy for one step do worse over many steps at the edge than a constant step [24], while [5] argue that the oscillation along v_1 acts as a penalty on sharpness. Eq. (15) is a statement about one step with the features fixed, not about the efficiency of gradient descent over training.

What holds λ_1 at $2/\eta$: self-stabilization, measured at the centre. Write the iterates as θ_j and the centre of the oscillation as the midpoint $c_j = (\theta_j + \theta_{j+1})/2$; each iterate lies a distance $\delta_j = \frac{1}{2}\eta\sqrt{\lambda_1}\,|x_j|$ from c_j along the top direction, with $x_j = \langle v_1, E_j\rangle$. With $S = \lambda_1$, $\alpha = -\nabla S\cdot\nabla L$ and $\beta = |\nabla S|^2$ evaluated at the centre, the self-stabilization analysis of [6] predicts that the sharpness at the centre changes per step by about $\eta(\alpha - \frac{1}{2}\beta\delta^2)$, so that it turns over when δ^2 reaches the central-flow value $\sigma^2 = 2\alpha/\beta$ [5].

Measured. From the saved parameters of the $\sin 10x$ run at $\eta = 1/\lambda_1(0)$ at step 10000, 200 steps of gradient descent were replayed, with $\eta\lambda_1$ between 1.992 and 2.013.

- One step against the best step of the same length (Proposition 6): the gradient step keeps the fraction -1.00 of x_j; the best step keeps 0.01; the term $k = 1$ of Eq. (15) is 97% (median) of the gap. The fixed-feature prediction of the error after each step is accurate to 8×10^{-4} of the error (at most 2.5×10^{-3}), which is 38% of the change of the error per step.

- The sharpness at the centre follows the self-stabilization prediction: over two steps its change has mean $+5.21 \times 10^{-3}$ against a predicted $+5.49 \times 10^{-3}$, with correlation 1.000, median ratio 0.98, and a residual whose root mean square is 5% of that of the change.

- The squared half-amplitude δ^2 grows from 1.1×10^{-5} and crosses the central-flow value $\sigma^2 \approx 2.3 \times 10^{-5}$ near step 10150; at that point the sharpness at the centre turns over, with $\eta\lambda_1$ at the centre reaching 2.008 and then falling.

An earlier version of this draft split the one-step change of λ_1 at the iterates into a part along the oscillation and a Taylor remainder, and attributed the cap to the second derivative of λ_1 along the top direction. That split is not a mechanism: for a smooth λ_1, the first-order term evaluated at the iterates contains the same second-derivative term with the opposite sign, and in the exactly solvable model of [14] the remainder has the opposite (destabilizing) sign. The attribution is withdrawn. The cap in these runs is the self-stabilization of [6].

7 Prior work and what is new

The following was checked against the literature up to October 2026.

Known.

- Rougher targets sharpen more and reach the edge of stability, smooth ones barely sharpen: [4] (Appendix L.2, one-dimensional tanh regression of Chebyshev polynomials of increasing degree), [9] (images with more high-frequency content), [27] (a "dataset difficulty" that weights label components on small singular directions), [17] (labels on bottom eigenvectors of the data Gram matrix).

- The output-layer norm grows at the rate $2\langle \hat{Y}, Y - \hat{Y} \rangle$ and sharpness tracks it: [17], as an assumption checked in experiments, with a proof for two-layer linear networks; exact versions in toy models [14, 12, 16, 11].

- The fixed-kernel formula $\langle \hat{Y}, Y - \hat{Y} \rangle = \sum_k s_k(1 - s_k)y_k^2$ (Proposition 5(a)): [17], Lemma 3.1.

- The homogeneity defect $\psi(z) = z\sigma'(z) - \sigma(z)$ in the dynamics of layer norms: [23] (weights, no biases).

- The general sharpening rate as a contraction of the kernel evolution: [10, 1, 6].

- The cap: self-stabilization [6] and central flows [5]; period-two orbits stabilized by higher-order terms [3, 28, 18, 15, 8].

- The part of the error off the top eigendirections decreases monotonically through loss spikes: [29].

- The best step of a given length is the trust-region (Levenberg–Marquardt) step, and its linearized residual is exactly $E_\gamma = \gamma(K + \gamma I)^{-1}E$ with $K = TT^*$, T the derivative of the forward map: [30], Eqs. (1.2) and (2.5), where the parameter is chosen so that $\|E_\gamma\| = \rho\|E\|$ (the same family of steps, indexed by residual level instead of step length). The edge of stability as the instability of an explicit scheme [26]; the top-subspace part of the update makes no progress [25].

- Damped Gauss–Newton (Fisher) preconditioning bounds the effective sharpness $\lambda_{\max}((F + \gamma I)^{-1}H)$ by $1 + (\varepsilon + \delta)/\mu_{\min}(F + \gamma I)$, with $\varepsilon = \|H - G\|$ and $\delta = \|G - F\|$, so damped second-order methods stay below their stability threshold [31]. Checked from the authors' code and README; the paper itself was not accessible.

- Stability bounds on the roughness of the learned function scale as $1/\eta$ at stable minima [21, 22].

Not found in the literature, so apparently new.

- The exact identity Eq. (13) for any network with a linear output layer, with the coefficient $\lambda_1 - k_L$ (the kernel of the hidden layers along v_1) and the remainder T_L; it turns the assumption of [17] into an identity.

- Lemma 1(b),(c): the defect δ_l with biases at the level of the output, and $\theta_l \cdot h_l = \lambda_1 - k_l + \varepsilon_l$.

- The measured decomposition: away from the edge, eigendirections with $\lambda_k < 10^{-3}\lambda_1$ carry 93–100% of the first-order sharpening, and q lies close to $\hat{Y}$.

- The anisotropic growth of the kernel: for $\sin 10x$ and $\sin 20x$ its 10th eigenvalue grows by 170 to 2200 and its 20th by 4×10^3 to 8×10^5 while λ_1 grows by 2 to 4 and stops at $2/\eta$ (eigenvalues below $10^{-12}\lambda_1$ are not resolved and are not used). At equal ηt this growth is the same at both step sizes; the edge caps λ_1 but does not redirect the growth. [13] report all eigenvalues growing; [2] report the opposite anisotropy in classification.

- The comparison of each gradient step at the edge with the best step of the same length, split by eigendirection, and the link to the duality of main.tex (the best step as the supremum of roughness-function bounds, computed by the Krylov series). The large share on v_1 follows from $\eta\lambda_1 \approx 2$, so this is a quantification, not a new phenomenon.

Withdrawn. The attribution of the cap to the second derivative of λ_1 along the top direction (Section 6); and a roughness-function form of Proposition 4, which is never stronger than the norm form.

8 What is proved, what is measured

Proved, with all terms kept, for any continuously differentiable activation. Proposition 1 (loss Hessian and kernel); Proposition 2 (one step of gradient descent, with remainder); Proposition 3 (gradient flow); Proposition 6 (one step against the best step of the same length, features fixed); Lemma 1, Theorem 1 and Proposition 4 (layer scaling, including the exact output-layer form Eq. (13)).

Proved with the kernel fixed. Proposition 5.

Open. (i) A bound on T in Eq. (11); the measurements below show that it is not small. (ii) Positivity of $\langle \hat{Y}, Y - \hat{Y} \rangle$ on the slow directions while the kernel moves. (iii) Whether the self-stabilization balance holds over longer replays and in the other runs that reach the edge.

Measured. Two hidden tanh layers of width 256 with biases, on 256 inputs, targets $\sin \omega x$ for $\omega \in \{1, 2, 5, 10, 20, 40\}$, full-batch gradient descent for 20000 steps at $\eta = 0.5/\lambda_1(0)$ and $\eta = 1/\lambda_1(0)$, with $\lambda_1(0) = 43.7$; the decompositions are evaluated every 2000 steps (132 checkpoints).

- $\lambda_{\max}(\nabla^2 L)$ and $\lambda_1(K)$ agree to within 1% on the second half of every run.
- $\eta\lambda_1$ reaches 2 for $\sin 10x$ at both step sizes and for $\sin 20x$ and $\sin 40x$ at the larger one; it stays at 0.51 to 1.05 for $\sin x$ and $\sin 2x$.
- The first-order term of Proposition 2 matches the exact one-step change of λ_1 to four digits away from the edge and to 1–5% at the edge; at initialization, where the output layer is zero, it vanishes and the whole first-step change is the remainder R.
- Away from the edge, eigendirections with $\lambda_k < 10^{-3}\lambda_1$ carry 93–100% of the first-order change, which is positive at almost every checkpoint; the change per step at step 10000 rises from 9×10^{-6} for $\sin x$ to 1.3×10^{-3} for $\sin 40x$ (smaller step size).
- Lemma 1: $\delta_L = 0$ and $\varepsilon_L = 0$ to rounding. For the hidden layers, $\|\delta_l\|$ is 0.2 to 2.7 times $\|\hat{Y}\|$, almost all of it from ψ (the bias part is at most $0.36\|\hat{Y}\|$, median $0.02\|\hat{Y}\|$), and $\theta_l \cdot h_l$ is about $0.1\lambda_1$ for the first layer and $-0.3\lambda_1$ to $0.5\lambda_1$ for the second, against $\lambda_1 - k_l$ for a homogeneous network. The part of the kernel of the hidden layers along v_1, $\lambda_1 - k_L$, has median $0.37\lambda_1$.
- Theorem 1, away from the edge ($\eta\lambda_1 < 1.9$, 89 checkpoints), as fractions of $-2\langle q, E \rangle$: the term $2\alpha\langle \hat{Y}, Y - \hat{Y} \rangle$ has median 1.37 (1.2 to 2.8 except where the total is near zero), D has median 0.00 (at most 0.05 in size except in one case), and T has median -0.36 (down to -1.75). The output-layer term has the right sign and order of magnitude, and T opposes it. At the edge, T carries all of the change.
- The function q lies close to the output: the cosine between q and $\hat{Y}$ is 0.95 to 1.00 for $\omega \leq 5$ and 0.6 to 0.8 for $\sin 40x$, and its cosine with $-E$ on the slow directions is about 0.02 for $\sin 40x$.
- The kernel grows mostly on its small eigenvalues: for $\sin 10x$ and $\sin 20x$ the 10th eigenvalue grows by 170 to 2200 and the 20th by 4×10^3 to 8×10^5, while λ_1 grows by 2 to 4 before it stops at $2/\eta$. (An earlier version quoted the 50th eigenvalue, which at initialization is below the resolution $10^{-12}\lambda_1$ of the computation.)
- At the edge: see Section 6.

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